Original Article

A Study on Consumer Awareness and Disposal Practice of E-waste and Their Determinants in Kolkata, West Bengal

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Abstract
Background: The growing burden of e-waste is an emerging public health concern. Consumers are important stakeholders, and percolating correct knowledge and generating awareness of e-waste are critical for proper e-waste handling and management, to minimize hazards and guide e-waste into a circular economy. This study aimed to assess consumer awareness and disposal practices at the individual level and their determinants. **Methods:** This online survey was conducted through a Google Form comprising a pre-designed, pre-tested structured questionnaire on e-waste-related awareness and disposal practices among adult residents of Kolkata. A total of 439 respondents submitted their responses from September to October 2020. Multivariable logistic regression was used to identify factors associated with satisfactory awareness levels and correct disposal practices. **Results:** A satisfactory level of awareness was found in 79% of participants. Younger age ($\leq$35 years vs. $>$35 years, AOR: 3.62), higher educational qualification (graduate and/or above vs. higher secondary or below, AOR: 3.73), and professional employment (vs. homemakers, AOR: 2.66) were associated with significantly higher awareness. Selling or exchanging devices during new purchases was the predominant disposal method for old electronics; only 4.1% deposited e-waste at designated e-waste collection points. **Conclusion:** Despite a satisfactory level of awareness, very few respondents disposed of old and unused e-products at e-collection centres. Generating awareness through mass media, introducing attractive take-back schemes, engaging municipal bodies for door-to-door collection, and building more e-collection centres are important measures for proper e-waste management.

Key Words: Consumer awareness, Disposal, Electronic waste, E-waste, WEEE

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I. Introduction

The growing burden of electrical and electronic waste (e-waste) is an emerging concern of this century[1,2]. E-waste is defined as items of electrical and electronic equipment (EEE) and their parts that have been discarded by the owner without intent of re-use, regardless of whether the item is still functional[1]. The Global E-waste Monitor 2020 reported 53.6 million tonnes of e-waste generated globally in 2019, of which Asia contributed 24.8 million tonnes; the global burden is projected to reach 74.7 million tonnes by 2030, yet only 17.4% was formally collected and recycled in 2019[3]. E-waste is a complex mixture of hazardous materials (heavy metals, brominated flame retardants, polyaromatic hydrocarbons) and recoverable precious elements[2,4-8]. In low- and middle-income countries, e-waste is often recycled crudely in household settings, exposing the general population — particularly women and children — to hazardous effects[7-11].

India is the third-largest e-waste generator globally, with only about a quarter of its waste stream handled through formal recycling capacity[3], and roughly 16% of national e-waste is generated by individual households[16]. Lack of awareness, weak enforcement of e-waste law and policy, and the absence of an organized collection system are identified as key barriers to proper e-waste management in India[4,7].

Kolkata, among the top ten contributors to India's e-waste stream and a major IT and business hub of the eastern zone with a population of over 14 million[10], is expected to see this burden grow further in the post-pandemic period. Against this backdrop, the current study aimed to assess consumer awareness and disposal practice of e-waste among residents of Kolkata, West Bengal.

II. Materials and Methods

This was an online, cross-sectional survey conducted via a pre-designed, pre-tested, structured Google Form questionnaire covering e-waste-related awareness and disposal practices. Adult residents of Kolkata were the target population. A total of 439 adult residents of Kolkata submitted valid responses between September and October 2020. Multivariable logistic regression was used to identify factors independently associated with (a) a satisfactory awareness level and (b) correct disposal practice.

III. Results

Of 439 respondents, the majority correctly associated multiple categories of items with e-waste: 61.7% identified irreparably damaged gadgets, functioning-but-unwanted devices, and broken electronic parts collectively as e-waste. Awareness that e-waste is hazardous to both humans and the environment was reported by 56.5%, while 5.7% did not know whether e-waste was hazardous at all. Most respondents (73.1%) believed e-waste needs special treatment before disposal, and 59.5% correctly identified that e-waste contains precious recoverable elements. Awareness of formal infrastructure and policy was markedly lower: only 12.5% knew of an e-waste collection centre in their city, and just 6.8% were aware of any e-waste-related law in India. **Table 1**

Fifteen participants (3.4%) scored the maximum on the awareness scale, and 347 (79%) scored ≥ 5, meeting the threshold for satisfactory awareness. In multivariable logistic regression, younger individuals (≤ 35 years vs. >35 years, AOR: 3.62; 95% CI: 2.05–6.39), higher educational qualification (graduate and/or above vs. higher secondary or below, AOR: 3.73; 95% CI: 2.05–6.78), and professional jobholders (vs. homemakers, AOR: 2.66; 95% CI: 1.05–6.75) were found to have significantly higher awareness. **Table 2** The non-significant Hosmer-Lemeshow test (0.312) indicated good model fit, with 9.4–14.6% of the variation in satisfactory awareness explained by the model (Cox & Snell $R^2 = 0.094$, Nagelkerke $R^2 = 0.146$).

About half of the respondents usually sell or exchange old or unused devices when buying new products, while 20% store unused gadgets at home; only 18 participants (4.1%) deposited e-waste at e-waste collection points. None of the sociodemographic factors examined were associated with correct disposal practice in this study ($p>0.05$). **Table 3** A large proportion (322, 73%) considered that financial responsibility for e-waste recycling and disposal should be shared among government, producers, traders, and citizens. Most respondents (356, 81.1%) reported rarely or never encountering e-waste-related advisories in print, digital, or social media, and salespersons never shared such information with 376 (85.6%) of respondents at the point of purchase. Only about a quarter of participants considered their own e-waste knowledge satisfactory, and about 90% (n=394) expressed a need for further education on the subject.

IV. Discussion

The current study showed that 347 (79.0%) of 439 participants had a satisfactory awareness level (total score ≥ 5). This is considerably higher than previously reported figures from other parts of India: Sivathanu et al.[4] reported 58.5% awareness among consumers in Pune; a 2013 Lucknow study among young adults reported 60.8% and 37.5% with medium and high knowledge respectively[11]; and a facility-based study among undergraduate medical students in Delhi found 77.3% aware, with 42.7% having adequate knowledge[17]. Miner et al.[9] reported 67.5% awareness in a 2018 Nigerian household survey.

In the present study, 61.7% considered broken, damaged, or simply unused electronic products collectively as e-waste. Around 4% considered e-waste non-hazardous and 5.7% did not know of any hazardous effect, while 19.6% considered it hazardous without knowing the specific effect. Shah et al.[7] in Ahmedabad similarly found 28% did not perceive e-waste as hazardous, and 37% considered it hazardous without being able to specify the effects; Miner et al.[9] found 68% aware of toxic effects, with 60.5% and 68% perceiving harm to health and environment respectively. About 73.3% of our participants felt e-waste needs special treatment before disposal, higher than the 50.9% reported by Miner et al.[9]

A majority of our participants (87.5% and 93.2% respectively) were unaware of any e-waste collection centre in Kolkata or any e-waste-related law or policy in India. Shah et al.[7] similarly reported only 37% aware of formal e-collection points and 11% aware of any government policy — despite more than six years having passed since that study, and despite e-waste handling and management law having been in place in India since 2011, awareness of formal infrastructure and policy remains low even among a better-educated study population such as ours.

Younger age, higher educational qualification, and professional employment were associated with better awareness in multivariable analysis, consistent with findings reported in previous studies[6,7,11,18]. Respondents who encountered e-waste advisories in media, or who were informed by salespersons at purchase, tended to have better awareness than those who rarely or never did. Only 4% of respondents used formal e-collection centres, while 19.4% still store old gadgets at home — broadly consistent with other Indian studies reporting 13–26% home storage[6,7] and around 18% formal disposal[7]. Nearly half sell or exchange old devices during new purchases, suggesting that consumers perceive continued monetary value in old electronics and are reluctant to discard them without compensation. Participants used their last mobile phone, PC, and TV for a median of 3, 4, and 8 years respectively, with access to newer technology cited as the predominant reason for replacement[7]. A significant proportion (73%) felt financial responsibility for e-waste recycling should be shared among citizens, producers/dealers, and government equally, though a wider community-based survey across all socioeconomic strata would be needed to confirm this finding more broadly.

Limitation

Being an online survey with voluntary, non-probabilistic sampling, this study could only capture the awareness and disposal practice of people with internet access and generally better educational status, which may explain the relatively high awareness observed. To minimise non-response, the number of questions was kept to a minimum, leaving several domains unexplored.

Item	n (%)
What counts as e-waste (select all that apply)	
Irreparably damaged gadgets	114 (26.0)
Unwanted but functioning devices	13 (3.0)
Broken electronic parts	7 (1.6)
All of the above	271 (61.7)
Don't know/not sure	34 (7.7)
Perceived hazard of e-waste	
Hazardous, specific effect unknown	86 (19.6)
Hazardous to humans	27 (6.1)
Hazardous to environment	36 (8.2)
Hazardous to both	248 (56.5)
Not hazardous	17 (3.9)
Don't know/not sure	25 (5.7)
Needs special treatment before disposal	
Yes	321 (73.1)
No	13 (3.0)
Don't know/not sure	105 (23.9)
Contains precious/recoverable elements	
Yes	261 (59.5)
No	31 (7.1)
Don't know/not sure	147 (33.5)
Aware of an e-collection centre in their city	
Yes	55 (12.5)
No	384 (87.5)
Aware of e-waste law in India	
Yes	30 (6.8)
No	409 (93.2)

Table 1: *Distribution of study participants according to level of awareness regarding e-waste (n=439)*

Parameter	Satisfactory awareness, n (%)	AOR (95% CI)	p
Age			
≤35 y (n=207)	183 (88.4)	3.62 (2.05–6.39)	<0.001
>35 y (n=232)	164 (70.7)	1 (Ref)	
Education			
Higher sec. or below (n=54)	30 (55.6)	1 (Ref)	
Graduate+ (n=385)	317 (82.3)	3.73 (2.05–6.78)	0.010
Occupation			
Professional (n=134)	119 (88.8)	2.66 (1.05–6.75)	0.040
Homemaker (n=31)	20 (64.5)	1 (Ref)	

Table 2: *Factors significantly associated with satisfactory awareness (non-significant covariates omitted for space; N=439)*

V. CONCLUSION

A significant proportion of study participants had satisfactory awareness regarding e-waste. However, correct disposal practice remains low and is driven mainly by exchange schemes rather than formal e-waste collection. Very few respondents were aware of current e-waste policy or the location of collection points in Kolkata. Building consumer awareness through social, digital, and print media, and at the point of purchase; motivating household-level segregation; introducing door-to-door collection; promoting take-back policies for all e-waste categories; and expanding e-collection infrastructure are all needed to address the growing burden of e-waste.

Parameter	Correct disposal, n (%)	OR (95% CI)	p
Age			
≤35 y (n=207)	152 (73.4)	1 (Ref)	
>35 y (n=232)	174 (75.0)	1.09 (0.71–1.67)	0.486
Gender			
Male (n=292)	213 (72.9)	1 (Ref)	
Female (n=147)	113 (76.9)	1.23 (0.78–1.96)	0.375
Education			
Higher sec. or below (n=54)	38 (70.4)	1 (Ref)	
Graduate+ (n=385)	288 (74.8)	1.25 (0.67–2.34)	0.486

Table 3: *Factors associated with correct disposal practice — none reached statistical significance (full occupation-category breakdown in original dataset, N=439)*

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Original Article

Impact of Early Management in Osteoarticular Infections in Infants and Neonates: A Two-Year Retrospective Study

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Abstract
Background: Osteoarticular infections in neonates and infants, including osteomyelitis and septic arthritis, can cause irreversible joint damage, growth disturbance, deformity, and functional disability. Early recognition and definitive institutional treatment are therefore important. This study evaluated the clinical characteristics, risk factors, causative organisms, treatment, and relationship between delay in institutional presentation and outcome. **Methods:** This retrospective study included 40 eligible neonates and infants treated between June 2019 and May 2021. Data were obtained from the institutional Hosp Gestor database after Institutional Review Board approval and informed consent. Delay in institutional presentation was categorized as 0–7 days, 8–30 days, 1–3 months, and >3 months. Demographic characteristics, infection type and site, risk factors, laboratory findings, microbiology, treatment, and sequelae were analyzed. Statistical analysis was performed using SPSS 27.0 and GraphPad Prism 5; chi-square or Fisher's exact test was planned for categorical comparisons, with $P \leq 0.05$ considered statistically significant. **Results:** Of 40 patients, 8 (20%) were neonates (<28 days). Males constituted 70% of the cohort and the mean age was 5.4 months. Septic arthritis was the most common diagnosis (53%), followed by osteomyelitis (43%) and concomitant septic arthritis with osteomyelitis (4%). The hip was the most frequent site of septic arthritis (76.2%). Seventeen patients (42.5%) had a history of previous infection or trauma; pulmonary disease accounted for 9 of these (52.9%). Atypical associated factors included preterm delivery/low birth weight, congenital talipes equinovarus, intramuscular injection, scabies, and bacterial conjunctivitis. Eleven patients (27.5%) had received treatment elsewhere, though details of agents and duration were unavailable. *Staphylococcus aureus* was the commonest isolated organism (35%); *Acinetobacter haemolyticus* was isolated in 2.5%. Surgical cleaning/arthrotomy and drainage was the commonest procedure. Institutional delay was categorized as 0–7 days (19 patients), 8–30 days (10), 1–3 months (8), and >3 months (3). The most severe documented sequelae occurred with prolonged delay, including epiphyseal growth arrest, hip dysplasia, chronic osteomyelitis with recurrent pus formation, and gap non-union; category-wise complication counts were not available in the available dataset, so a valid inferential p-value for delay versus outcome could not be calculated. **Conclusion:** The most severe documented sequelae occurred among patients with prolonged delay in institutional presentation. Because category-wise complication counts were unavailable, the observed pattern should be interpreted descriptively rather than as a statistically confirmed association. The findings support prompt recognition and referral of neonatal and infant osteoarticular infections, particularly in settings where delayed presentation and treatment by informal health practitioners may occur.

Key Words: Infants, Neonates, Osteoarticular infection, *Staphylococcus aureus*, Time-to-treatment, Treatment outcome